

Grade 7 Accelerated
Unit 2
Lesson 4 Homework

Name _____
Date _____
Period _____

1. Find the square root of each of the following.

$$\sqrt{16} =$$

$$\sqrt{-16} =$$

2. Find the cube root of each.

$$\sqrt[3]{27} =$$

$$\sqrt[3]{-27} =$$

3. Which value from questions 1 and 2 was not a real solution? When does a square or cube root result in a non-real solution?

Consider $\sqrt{23}$, an irrational number.

4. Between which two perfect squares does 23 lie?
5. Between which two integers is the value of $\sqrt{23}$?

Now consider $\sqrt[3]{55}$, another irrational number.

6. Between which two perfect cubes does 55 lie?
7. Between which two integers is the value of $\sqrt[3]{55}$?

8. Use the approximation of $\sqrt{78}$ to approximate the value of each expression in the table.

Expression	Approximate Value
$\frac{\sqrt{78}}{4}$	
$-3\sqrt{78}$	
$\sqrt{78} + 5$	

9. Approximate the value of each expression.

	Expression	Approximate Value
A	$\sqrt[3]{120}$	
B	$\sqrt[3]{120} - 2$	
C	2π	
D	$1 + \pi$	

10. Plot and label the approximate location of each expression from question 9 using letters **A**, **B**, **C**, and **D** on the number line.

